CERTIFICATE

*This is to certify that Mr./Ms.*____

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Exam No. ______________________ *has satisfactorily completed his / her term*

work in ____ *for*

the term ending __________________ *in 20*___ */ 20*___.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Sign of the Faculty *Head of the Department*

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

Department of Civil Engineering

CL 408: IRRIGATION & HYDRAULIC STRUCTURES

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TUTORIAL 1

IRRIGATION & IRRIGATION METHODS

1. Define irrigation and explain the necessity of irrigation for a tropical country like India.
2. What are the advantages and ill effects of irrigation?
3. What is meant by surface and subsurface irrigation?
4. Discuss the quality standards required for irrigation water.
5. Write a short note on sub-surface irrigation, state clearly the conditions under which this method is suitable.
6. Explain the various methods of 'surface irrigation system', bringing out clearly merits and applicability of each method.
7. Explain sprinkler and drip irrigation methods stating clearly the component parts, merits and demerits of them.
8. Briefly explain about the following :
    a. Intensity of irrigation
    b. Lift irrigation
    c. Tank irrigation

Date:

TUTORIAL 2

WATER REQUIREMENT FOR CROPS AND IRRIGATION EFFICIENCIES

Q - 1

A field channel has a culturable command area of 2000 Ha. The intensity of irrigation for gram id 30% and for wheat is 50%. Gram has a core period of 18 days and kor depth of 12cms while wheat has a kor period of 15 days and kor depth of 15 cms. Calculate the discharge of field channel.

Q -2

A water course command area of 800 Ha. The intensity of irrigation of paddy crop is 50 % . The transplantation of paddy crop takes place 15 days and total depth od water required by crop is 60 cmson the field during the transplation period, given that the rain falling on the field is 15 cms. Find the duty of the irrigation water for the crop on the field during transplantation, at the head of distributor, assuming water losses 20% in the water course. Also calculate the discharge required for the water course.

Q -3

In the following table, the data regarding the canal irrigation are depicted.

Sr. NO.	**Crops**	**Crop period, days**	**Area , hactares**	**Duty at the head of the canal Ha/cumecs**
1	Sugarcane	320	720	575
2	Sugarcane (overlap)	120	125	575
3	Paddy	122	430	650
4	Spiked Millet	114	1025	2100
5	Wheat	120	830	1580
6	Great Millet	110	570	1620
7	Vegetables	105	350	625

Find the discharges required at the head of the main canal taking time factor as 0.80 and canal losses as 15% . Take capacity factor as 0.75. Determine the gross storage capacity of the reservoir. Consider 10% as dead storage.

Develop the computer code for determination of canal discharge and gross storage requirement of reservoir.

Q-4

A distributory canal supplies water at the rate of 2.15 m^3/s to the field and at the Parshall flume measures the flow at the rate of 1810 litre per seconds. An area of 1500 Hactares are to be irrigated within stipulated time of 9 Hrs. From that area the runoff loss is considered as 7330 m^3. The water penetrating depth at the head of the field is 1.70 m and at the tail end of the field is 1.25m. Available moisture holding capacity of the field soil is 19 cms per metre of the depth. Application of water designed at the allowable moisture depletion 60% of the available moisture . The effective root zone depth is 1.90 m.

Determine (I) Water conveyance efficiency
(ii) Water application efficiency
(iii) Water storage efficiency
(iv) Water distribution efficiency

Develop the computer code for the determination of all efficiencies mentioned above.

Q-5 Objective type Questions

1.1 The available moisture holding capacity of the soil is 13 cm per m depth of soil. If a crop with a root zone 0.80 m and consumptive use of 5mm/day is to be grown, the frequienciy of irrigation for restricting the moisture depletion to 50% of available moisture is

(a) 6.5 days (b) 26 days (c) 13 days (d) 10 days

1.2 Consumptive use is

(a) the water used up in the plant metabolism

(b) the sum of evapotranspiration and the amount used up in plant metabolism.

(c) the sum of evpotranspiration and infiltration losses.

(d) the combined use of surface and groundwater resources.

1.3 For the irrigation of a crop , the base period B in days and delta Δ in meters to the duty D in ha /(m^3/s) at the field as

(a) D = [0.864 B/A] (b) D = [8.64 B/A] (c) [0.864 Δ/B] (d) Δ = [8.64 D/B]

1.4 A canal water is to irrigate 1200 Ha of land growing rice of 140 days base period and having delta of 134 cms. If the canal water is used to irrigate wheat of base period 120 days and having a delta of 52 cms, then the area that can be irrigated is

(a) 2651 ha. (b) 543 ha. (c)3608 ha. (d) 399ha.

1.5 In a certain irrigation project , in a year, 72% and 66 % of the culturable command area in Kharif and Rabi respectively and remain without water. The intensity of irrigation in that year of that project was

(a) 31 % (b) 69 % (c) 138 % (d) 62 %

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TUTORIAL 3

Stability Analysis of an Earth Dam

For the zone type dam section determine the stability for u/s slope and d/s slope for sudden draw down condition and steady seepage conditions respectively by Fellenius method by incorporating with and without seismic conditions.

Geometric Details:-

U/s slope – Core	= 1:1
D/s slope – Core	= 1:1
U/s slope – Shell	= 2.5:1
D/s slope - Shell	= 2:1
Top width	=
Reservoir Depth	=
Free Board	=

Geotechnical Details:

Particulars	Core	Shell
Specific Gravity	2.50	2.60
Porosity	33%	20%
Cohesion, kN/m^2	38	5.00
Angle of Internal Friction	20^0	33^0

Take modulus of elasticity E = 2500 kN/m^2 and density of soil = 2620 kg/m^3. Consider 10% damping rate for seismic vibration for getting response curve.

Q -2 Objective questions

2.1 Whether the following statements are true or false

(a) The u/s face of an earth dam is considered as equipotential line.

(b) The pressure above the phreatic line is negative hydrostatic pressure.

(c) To prevent the seepage through the pervious strata of foundation the cut of trench provided and the material used in that will be of shell.

(d) Rip – rap is provided at the d/s face of the dam to prevent the erosion of d/s slope.

(e) For steady state seepage condition stability checking the reservoir must be full.

2.2 The stability of d/ s slope of an earth dam is checked for

(a) sudden drawdown (b) steady seepage

(c) construction period (d) against the gully development

2.3 The cutoff is provided in the earth dam to

(a) reduce the seepage (b) prevent piping

(c) prevent erosion of slopes (d) sloughing at the toe

2.4 The pressure above the phreatic line is

(a) positive hydrostatic. (b) negative hydrostatic

(c) atmospheric. (d) vacuum.

2.5 For analyzing stability of dam in obtaining the resisting force below phreatic line the

(a) unit dry weight of soil to be taken

(b) unit moist weight of soil to be considered

(c) unit saturated weight of soil to be taken

(d) unit submerged weight of soil to be considered

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TUTORIAL 4

Design and Stability analysis of Gravity dam

For the particular dam site, the gravity dam is proposed. Design the gravity dam for which the specifications are listed as under.

1. Reservoir depth, m = 70 m
2. Tail water depth, m = 10m
3. Unit weight of Concrete = 24kN/m^2
4. Unit weight of water =10kN/m^2
5. Drainage holes are located at 2m Away from the vertical axis of the dam where 70% of total uplift pressure head is dissipated
6. Horizontal Seismic Coefficient $ά_H$ = 0.1g
7. Vertical seismic Coefficient $ά_V$ = 0.05g
8. Wind velocity = 84 km/hr
9. Fetch of the reservoir = 60 kms

Check the stresses (Normal, Principal and Shear) at heel and at toe of the dam for reservoir full and reservoir empty conditions when drain is operative. Also determine the stability of dam against
(a) Overturning
(b) sliding and
(c) shear action

Prepare drawing in detail for gravity dam designed and show all details.

Objective Questions

3.1 The axis of dam is the

(a) line joining the mid point of the base.

(b) centre line of top width of dam.

(c) line of the crown of the dam on the d/s side

(d) line of the crown of the dam on the u/s side.

3.2 The centre pf pressure of wave pressure due to wave of height h_w acting on a gravity dam will be at height above the maximum still water level of

(a) $h_w/2$ (b) $3h_w/8$ (c) $h_w/3$ (d) $2h_w/3$

3.3 At a joint in a gravity dam of base width 140m, the sum of vertical forces and the sum of horizontal forces are 180 MN and 120 MN , respectively. The coefficient of friction is 0.7 and shear strength of concrete is 3.2 MPa . The shear friction factor of safety of this joint is

(a) 1.05 (b) 0.95 (c) 4.78 (d) 3.73

3.4 At the base of the gravity dam the vertical stress at the toe was 2.0 MPa. If the d/s face has the slope of 0.707:1 and there is no tail water , The principal stress at toe is

(a) 2.45 Mpa (b) 3.00 MPa (c) 2.00 MPa (d) 2.83 MPa

3.5 The maximum height of low gravity dam of elementary profile , to be built of concrete of relative density 2.4 on a foundation material of safe allowable stress 7.5 MPa by ignoring uplift force is

(a) 319 m (b) 484 m (c) 125 m (d) 225 m

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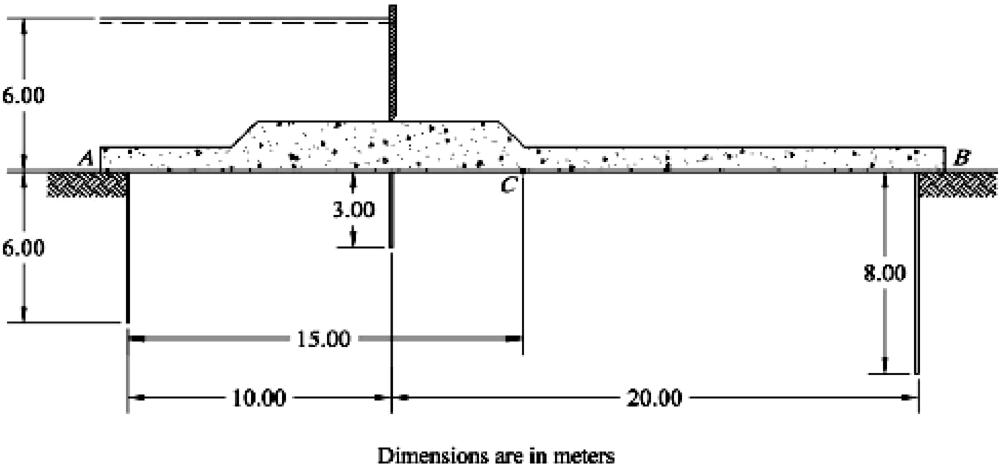
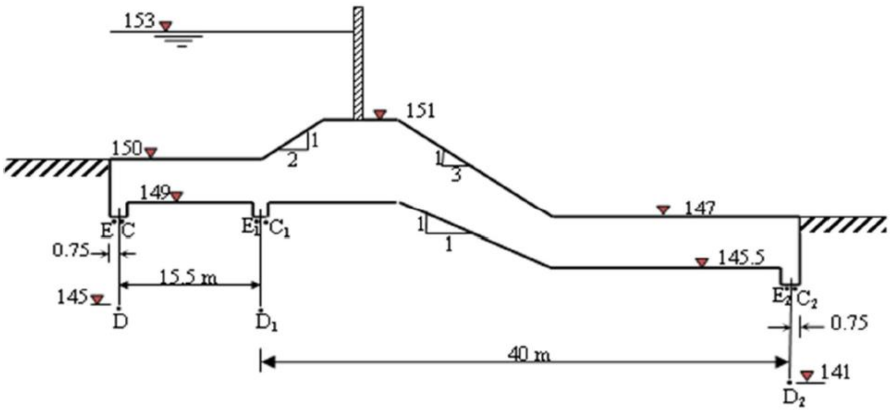
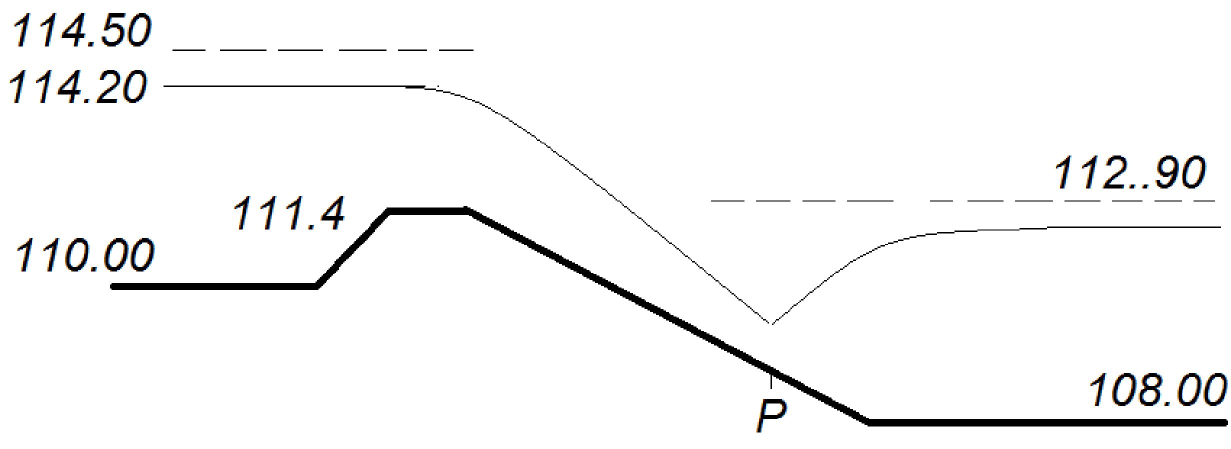
TUTORIAL 5

DIVERSION HEAD WORKS

Q-1	Find the hydraulic gradient and uplift pressure at a point 15 m from the upstream end of the floor in the figure below. Also find the floor thickness at C if specific weight of material is 2.24 
Q- 2	Determine the corrected percentage pressure at the key points by Khosla's theory. Also check the thickness provided and exit gradient. Safe exit gradient = 1/6 

Q -3 For the hydraulic structure constructed on pervious strata determine the location of point P where the hydraulic jump will be formed. Also draw the pre jump and post jump profile.

The intensity of discharge is 12 cumecs per m length



Q – 4 Objective Questions

4.1 For a given foundation geometry of barrage, the exit gradient is

(a) dependent on the porosity of foundation soil

(b) independent of the type of soil

(c) independent of the applied head

(d) dependent on the size of the foundation soil

4.2 A non cohesive soil has a porosity 30% and the relative density of the soil particle is 2.70. The value of critical exit gradient is

(a) 0.81 (b) 1.19 (c) 1.00 (d) 1.89

4.3 A weir consists of 36 m long horizontal floor with two sheet piles of 6m and 8 m depth at u/s and d/s end of the floor respectively. Under impounded depth of 4 m above the floor and with no tail water , the uplift pressure head at the mid point of the floor by Bligh's creep theory is

(a) 2.00m (b) 1.44m (c) 2.12 m (d) 1.88m

4.4 As a result of construction of diversion head works across a river there will be rise in the flood level on the u/s side of the structure and it is called as

(a) free board (b) uplift (c) aggradation (d) afflux

4.5 When a weir is constructed on alluvial river the retrogression of the d/s channel

(a) goes on up to maximum value and gradually recovers towards the original bed levels over a period of time.

(b) reaches a maximum value and gradually gets aggraded to flatter slope than the original value.

(c) continuous increase with time

(d) stabilizes after reaching a limit value

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TUTORIAL - 6

SPILLWAY AND ENERGY DISSIPATORS

Q – 1 An overflow spillway with u/s face vertical is to be designed for a flood of 2100 cumecs. The level of the spillway crest is at RL 130.00 and the river bed is RL 100.00 . The end walls of a spillway are 79 m apart and four round nose pier , each 1 m wide. Determine the total head over the crest and u/s and d/s profile of spillway. Also obtain water surface profile by using tables.

Q – 2 Design a suitable stilling basin at the toe of the spillway from the following data.

(1) Discharge intensity q = 24.6 cumecs / m

(2) HFL in reservoir = 100.00

(3) Spillway crest level = 95.00

(4) River bed level = 59.00

(5) Tail water level = 60.00

(6) coefficient of velocity = 0.90

Adopt the tail water depth 5% greater than the sequent depth. Assume L_b/y_2 = 4.3

Q -3 Objective Questions

5.1 If the head over an ogee spillway is less than the design head,

(a) the pressure on the spillway crest will be negative

(b) the coefficient of discharge will be less than the design coefficient of discharge.

(c) The coefficient discharge will be larger than the design coefficient od discharge.

(d) the cavitation phenomenon will occur.

5.2 The crest profile of an ogee spillway , downstream of the apex under the design head H_d can

be expressed as (y/H_d) = K ($x/H_d)^n$ where the coefficients K and n , for a vertical u/s spillway would , respectively, be

(a) 0.50 and 1.85 (b) 0.85 and 2.0 (c) 2 and 0.85 (d) 2.0 and 1.5

5.3 In a solid roller bucket type energy dissipater , the energy dissipation is

(a) due to formation of hydraulic jump

(b) partly due to the interaction of the free jet with air and partly due to the impact on d/s channel be.

(c) due to interaction of two complimentary rollers.

(d) partly due to the lateral spreading of jet and partly due to the interaction of two rolers.

5.4 If at the foolt of the spillway y2 is the sequent depth of a hydraulic jump that could form on a leveled floor , a roller type energy dissipater is used when the tail water depth is

(a) considerably more than y2

(b) considerably less than y2

(c) slightly more than y2

(d) slightly less than y2

5.5 It is required to control the maximum water surface of a small reservoir within narrow limits of the full reservoir level , the spillway that has the most favorable discharge characteristics for this purpose is

(a) over flow spillway (b) morning glory spillway

(c) siphon spillway (d) side – channel spillway

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TUTORIAL - 7

CANAL DESIGN

Q- 1 Design an unlined irrigation channel for alluvial soil by Kennedy's theory for following data:

Discharge = 45 cumecs, Channel bed slope = 1:5000, N = 0.0225, m = 1.05, Side slope of trapezoidal channel = ½ :1.

Q- 2 Design a channel section of an unlined irrigation channel for alluvial soil for given data by Lacey's regime theory:

Discharge (Q) = 30 cumecs, Silt Factor (f) = 1 and Side slope = ½ :1. Also calculate the longitudinal slope of the channel.

Q- 3 Design an unlined irrigation channel for non-alluvial soil using Chezy's formula for data given below:

Q = 45 cumecs, Mean velocity = 0.9 m/sec, B = 5D, Side slope = 1:1, K = 1.2.

Q- 4 Design an irrigation channel by Kennedy's theory using Garret's diagram for following data:

Full supply discharge = 7 cumecs, Rugosity co-efficient = 0.0225, Bed slope = 1/4444, Critical Velocity Ratio (C.V.R.) = 1 and Side slope of channel = ½ :1.

Q- 5 Design an irrigation channel by Lacey's regime theory using Lacey's regime diagram for following data:

Full supply discharge = 15 cumecs, Lacey's silt factor = 1.0 and Side slope of channel = ½ :1.

Q-6 Design a trapezoidal shaped concrete lined canal to carry a discharge of 100 cumecs at a slope of 25 cm / km. the side slope of channel is 1.5:1. The value of N is 0.016, limiting velocity = 1.5 m/sec.

Q – 7 Objective Questions

6.1 Garret's diagram for the stable channel is based on

(a) Kennedy's theory and Manning's formula

(b) Kennedy's theory and Kutter's formula

(c) Lacey's theory and Manning's formula

(d) Lacey's formulae.

6.2 When an alluvial channel attains it's regime it will have side slopes

(a) of value equals to the angle of repose of alluvium

(b) of value equals to the angle of repose of alluvium under water.

(c) of value 0.50 vertical : 1 Horizontal

(d) of value o.50 Horizontal : 1 Vertical

6.3 Two channels A and B , in different locations, have been designed on Lacey's regime theory to carry the same discharge Q. If the median size of the alluvium at the location of the channel A is coarser than at the location of channel B , then

(a) channel B is deeper

(b) channel A is deeper

(c) channel B will have larger wetted perimeter

(d) channel A will have smaller longitudinal slope.

6.4 Which of the following is not a disadvantage of cement concrete lining?

A cement concrete lining

(a) can get easily punctured by weed growth.

(b) develops frequent cracks due to contraction arising out of temperature change and drying

(c) develops crack due to the settlements of sub grade

(d) is likely to be damaged by alkaline water.

6.5 A counter berm is provided

(a) at the water side of the slope in filling to provide protection against rain water

(b) at the outer side of the filling to provide adequate cover over the phreatic line.

(c) in the cutting to provide quick silting of the sides to form a berm

(d) in the water side and at the ground level to make the bank line parallel.

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TUTORIAL 8

CROSS DRAINAGE WORKS, OUTLETS AND CANAL REGULATION WORKS

Q-1 Define cross drainage works and classify cross drainage work with neat sketches.

Q-2 Enlist different types of canal outlets.

Q-3 Define and enlist types of falls, HR, CR and canal escapes with neat sketches.

Canal Regulating Works

C.1 In a canal head regulator the maximum height of the gate opening is equal to the difference between the

(a) pond level and sill level of the regulator

(b) HFL and sill level of the regulator

(c) HFL of the river and FSL of the canal

(d) pond level and FSL of the canal

C. 2 A cross regulator is built

(a) below the head of an off taking channel at a suitable d/s location.

(b) in a main channel d/s of an off taking channel or escape

(c) along the axis of the head of the off take channel.

(d) at the u/s of an off take.

C.3 Cross regulators are utilized to

(a) increase the water level u/s of the structure when a main canal is running with low supplies.

(b) control the sediment entering into an offtake

(c) dispose excess of supply from the main channel into the drainage.

(d) control the silt load in the main channel.

C.4 In straight glacis type falls, the energy dissipation is essentially due to

(a) water cushion

(b) baffle and friction blocks only

(c) fluming action

(d) hydraulic jump assisted by baffle blocks

C.5 In Sarda type fall , the maximum uplift pressure on the floor is expected when the u/s water surface is

(a) at the crest level of the fall and d/s water level is at the brim of cistern.

(b) at the crest level of the fall and there is no water in the d/s channel

(c) at the bed level of the u/s channel and d/s water is at the brim of cistern

(d) at the full supply level and there is no flow in the d/s channel

Cross- Drainage Works

CD.1 A canal carries a discharge of 20 m^3 at a depth of 1.50 m and has it's bed 3.5 m higher than the bed of drainage it has to cross. If a drainage has a high flood depth of 2.50 m, the type of the cross drainage works appropriate to this site is

(a) aqueduct (b) siphon aqueduct (c) siphon (d) super passage

CD.2 In siphon aqueduct the worst condition uplift on the floor occurs when

(a) the canal and drainage are full

(b) the canal is full and the drainage empty and water table at the stream bed

(c) the canal is empty and drainage flow at HFL and the water table is at the HFL of the stream

(d) the canal is full and drain is empty

CD.3 A level crossing type of cross drainage work consists of

(a) one regulator only (b) two weirs only

(c) one weir and two regulators (d) two weirs and two regulators

CD.4 The following structure serves the purpose of ‘ safety Valve ‘ for a canal.

(a) head regulator (b) cross regulator (c) canal escape (d) canal fall

CD.5 The following data is available for cross drainage project

Item	Canal	Drainage
FSL/HFL	105.00	104.00
Bed Level	100.00	102.00
Discharge	80 m^3/s	12 m^3/s

The most appropriate cross drainage works for this situation is

(a) aqueduct (b) siphon aqueduct (c) siphon (d) super passage

Irrigation Outlets

IO.1 The ratio of the rate of change of discharge in an irrigation outlet to the rate of change of discharge in the distributor is known as

(a) sensitivity (b) modular ratio (c) flexibility (d) setting

IO.2 The outlet factor is

(a) the duty based on the discharge

(b) the ratio of mean discharge of outlet to the full supply discharge of the canal

(c) the inverse of the duty of the outlet

(d) the ratio of number of days the outlat is actually open to the number of days it was suppose to be open as per design

IO.3 Gibb's module is

(a) a flexible module (b) an adjustable proportional module

(c) rigid module (d) non-modular outlet

IO.4 By considering the channel index as 5/3 the setting of an orifice type irrigation outlet to have proportionality is

(a) 0.67 (b) 0.90 (c) 0.15 (d) 0.30

IO.5 A n irrigation out is proportional when it's

(a) Sensitivity =1

(b) flexibility = 1

(c) setting = 1

(d) channel index = 1

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TUTORIAL 9
RESERVOIRS

Q-1 Define: Reservoir. Discuss briefly different types of reservoirs and the purposes served by each one.

Q-2 Which are the factors affecting selection of site for a reservoir?

Q-3 Explain 'Storage zones' with neat sketch.

Q-4 Derive the equations of Area-elevation curve and Capacity-elevation curve of a reservoir site.

Q-5 Define following:

Catchment yield, Reservoir yield, Dependable yield & Safe yield

Q-6 The yearly data for a catchment of a proposed reservoir site for 30 years is given in following table. **Compute the dependable rainfall for 55% & 70% dependability.**

Year	**1945**	**1946**	**1947**	**1948**	**1949**	**1950**	**1951**	**1952**	**1953**	**1954**
Rainfall, cm	78	76	101	112	80	85	116	120	140	150
Year	**1955**	**1956**	**1957**	**1958**	**1959**	**1960**	**1961**	**1962**	**1963**	**1964**
Rainfall, cm	99	98	66	70	104	92	95	87	69	91
Year	**1965**	**1966**	**1967**	**1968**	**1969**	**1970**	**1971**	**1972**	**1973**	**1974**
Rainfall, cm	131	138	126	128	120	111	110	83	82	94

Q-7 **Calculate the capacity for given demand graphically and analytically:**

The following table gives the mean monthly flows in a river. Calculate the minimum storage required to maintain a demand rate of 40 m^3/sec for two months.

Month	**Jan**	**Feb**	**March**	**April**	**May**	**June**	**July**	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**
Mean Flow, m^3/sec	60	45	35	25	15	22	50	80	105	90	80	70

Q-8 **Calculate the demand for given capacity:**

What is the maximum uniform rate that can be maintained by storage of 3600 m^3/sec days for given data?

Month	**Jan**	**Feb**	**March**	**April**	**May**	**June**	**July**	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**
Mean Flow, m^3/sec	60	45	35	25	15	22	50	80	105	90	80	70

Q-9 Which measures have been taken to control the sedimentation before and after constructing the reservoir?

Q-10 A proposed reservoir has a capacity of 450 ha. m and catchment area 250 km^2. The annual streamflow averages 10 cm of runoff and production of annual sediment is 0.045 ha. m / km^2. Calculate the annual sediment inflow in the reservoir in Mm^3.

Multiple Choice Questions:

1. Which reservoirs are used to feed the canal systems for irrigation and power generation?

   a) Storage and conservation reservoirs

   b) Flood control reservoirs

   c) Distribution reservoirs

   d) Drought control reservoirs

2. The reservoirs for small storage and short period of time are

   a) Storage and conservation reservoirs

   b) Flood control reservoirs

   c) Distribution reservoirs

   d) Drought control reservoirs

3. The level up to which the reservoir shall be full of water is called

   a) High level

   b) Maximum level

   c) Pool level

   d) Surface level

4. What is the volume of water which is available between maximum reservoir level and normal pool level called?

   a) Useful storage

   b) Dead storage

   c) Surcharge storage

   d) Specific storage

5. Some amount of water can be stored beyond maximum pool level. State true or false.